# Final Exam (PowerBI)

## 1- Part 1: Data Loading

1.1 Load the following data tables into PowerBI (**Calendar, Inventory_Items, Item_Categories, Item_Subcategories, Return_Transactions, Sales_Orders, sales_Regions**)

1.2 Append **Clients_Part1, Clients_Part2** tables into a new table named "**Clients**"

## 2- Part 2: Data Manipulation

### 2.1 Calendar Table

A. Add new column for **start of the month**
B. Add new column for **start of the week**
C. Add new column for **name of the day**
D. Create a new column named **'Week End'** where the **value** is set to **'Weekend'** if the day is **Friday** or **Saturday**, and **'WorkDay'** for all **other days**.
E. Change format for calendar columns (**Date, start of the month, start of the week** to be as **01-Jan-18** format)

**Sample results:**

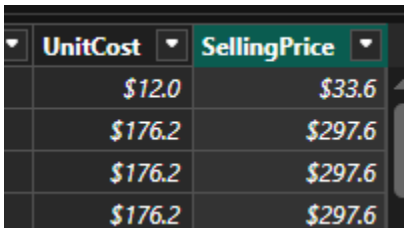
| Date | Start of Month | Start of Week | Day Name | Week End |
|---|---|---|---|---|
| 03-Jan-18 | 01-Jan-18 | 31-Dec-17 | Wednesday | Work Day |
| 04-Jan-18 | 01-Jan-18 | 31-Dec-17 | Thursday | Work Day |
| 05-Jan-18 | 01-Jan-18 | 31-Dec-17 | Friday | Week END |
| 06-Jan-18 | 01-Jan-18 | 31-Dec-17 | Saturday | Week END |

### 2.2 Clients table

A. Add new column named as **full name** that concatenate **First_name** & **FamilyName**
B. Create a new conditional column labeled **'Salary Level'** to categorize clients into tiers **based on** their **salary**, as follows
   - **Very High** if Salary >=120000
   - **High** if Annual Salary between (120000 – 80000)
   - **Mid** if Annual Salary between (80000 – 40000)
   - **Low** if Annual Salary < 40000

### 2.3 Inventory_items table

A. Change the format for **UnitCost** and **SellingPrice** as:
   - Currency
   - Number of decimals = 1



| UnitCost | SellingPrice |
|---|---|
| $12.0 | $33.6 |
| $176.2 | $297.6 |
| $176.2 | $297.6 |
| $176.2 | $297.6 |

### 2.4 Sales_Reigons table

A. Change Data Category for **ContinentName** to be Continent
B. Change Data Category for **CountryName** to be Country
C. Create hierarchy for ContinentName and CountryName




## 3- Part 3: Data Relations

Build the following relations between database Tables

| From Table (Column) | To Table (Column) |
|---|---|
| Inventory_items (Subcategory_ID) | Item_subcategories (Subcategory_ID) |
| Item_subcategories (Category_ID) | Item_categories (Category_ID) |
| Return_transactions (Product_ID) | Inventory_items (Product_ID) |
| Return_transactions (ReturnDate) | Calender (Date) |
| Return_transactions (Territory_ID) | Sales_Regions (Territory_ID) |
| Sales_orders (Client_ID) | Clients (Client_ID) |
| Sales_orders (OrderDate) | Calender (Date) |
| Sales_orders (Product_ID) | Inventory_items (Product_ID) |
| Sales_orders (Territory_ID) | Sales_Regions (Territory_ID) |

# 4- Part 4: DAX

Create a new table, call it **"ALL Measures"** and put all the following **DAX measures.**

## 4.1 Calculated column

A. Make a calculated column in **Sales_Orders Table** to get the **Unit Price** from **inventory_item** table.
B. Make a calculated column in sales orders table to get unit cost from inventory item table.

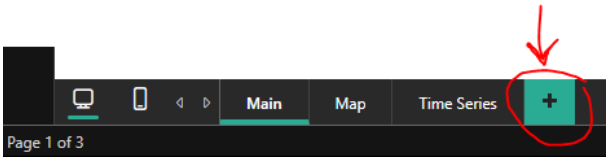
## 4.2 Calculated DAX Measure

A. Create a new measure named ***Total Revenue*** that calculates the **sum** of (**Order Quantity × Unit Price**) from the **Sales Orders table**.
B. Create a new measure named ***Total Cost*** that calculates the **sum** of (**Order Quantity × Unit Cost**) from the **Sales Orders** table.
C. Create a new measure named ***Total Orders*** to **count the number of distinct orders IDs** in the **Sales Orders** table.
D. Create a measure named ***Workday Orders*** to **count orders** that **occur on workdays** (Monday through Friday).
E. Create a measure named ***Weekend Orders*** to **count orders** placed on **weekends** (Saturday and Sunday).
F. Create a measure named ***Average Price*** to calculate the **average unit price** from the **Inventory Items** table
G. Create a measure named **% Weekend Orders** that calculates the **percentage** of **weekend orders** (Saturday and Sunday) relative **to total orders**.
H. Create a measure named ***% Workday Orders*** that calculates the **percentage** of **weekday orders** (Monday through Friday) **relative to total orders**
I. Create a measure named ***Total Returns*** that calculates the **sum of returned quantities** from the **Return Transaction table**
**J.** Create a measure named ***Return Percentage*** that calculates the **ratio of total returned** quantities **to all Return Quentities** in **return transaction table** and **formatted as a percentage %.**
K. Create a measure named ***Expensive Item Revenue*** that calculates the **total revenue for items with** $unit\ price >= average\ unit\ price$ in the Inventory Items table
L. Create a measure named ***Total Profit*** that calculates the **difference** between **Total Revenue and Total Cost.**
M. Create a measure named ***Previous Month Orders*** that calculates the **total orders** for the **previous month**

N. Create a measure named ***Previous Month Returns*** that calculates the **total returned** quantities for the **previous month**
O. Create a measure named ***Previous Month Revenue*** that calculates the **total revenue** for the **previous month**
P. Create a measure named ***Target Revenue*** that calculates **a 5% growth** target based on the **previous month's revenue** $(Previous\ Month\ Revenue \times 1.05)$
Q. Create a measure named ***Revenue Target Achieved*** that returns **"1"** when $Total\ Revenue >= Target\ Revenue$, and **"0" otherwise**.
R. Create a measure named **Year-To-date Profit** that calculates cumulative **profit from** the **start of the current year to the latest date** in the filter context.

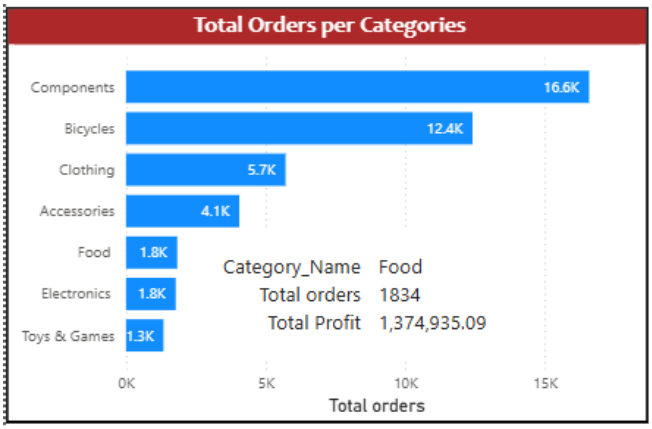
## 5- Part 5: Visualization & Dashboards

- Create these report pages: **Main**, **Map** and **Time Series**, **you can add new Page if needed**

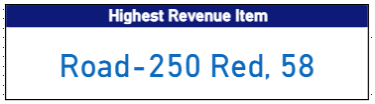



- **For all visuals make chart boarder (width = 2px)**

A. Use a **Logo** that Have **your name** and **student ID**
B. Create a **Stacked bar chart** with **title** "Total Orders per Category " – (title text color = white, background color Red)
   - Hide Y-axis title.
   - Data labels to be inside the bar.
   - Need to view total profit when I put the pass over any bar (tooltips)



C. Create **5 card visuals** with these configurations
   1- Total orders with Display units = None
   2- Total Profit with display units = Auto
   3- Total Returns with display units = None
   4- Total revenue with display units = Auto
   5- itemName for Inventory Items table that display the most Revenue items, use title name and background color as the below picture




D. **Matrix visualization** with the following
   - Rows: start of the month
   - Values: (Total Orders, Total Returns, Return Percentage, Total Revenue, Year-to-Date Profit)
   - Make the matrix format as below figure. (header, colors indicators for Year-to-Date Profit, Return Percentage)

| Start of Month | Total orders | Total Returns | Return % | Total revenue | Year to date Profit |
|---|---|---|---|---|---|
| 01-Jan-19 | 242 | 91 | 3.33% | 218,210.81 | 93,711.37 |
| 01-Feb-19 | 267 | 90 | 3.29% | 264,432.41 | 209,231.48 |
| 01-Mar-19 | 266 | 94 | 3.44% | 198,372.75 | 293,081.72 |
| 01-Apr-19 | 290 | 74 | 2.71% | 295,432.81 | 424,200.91 |
| 01-May-19 | 329 | 85 | 3.11% | 337,310.18 | 570,320.98 |
| 01-Jun-19 | 312 | 86 | 3.15% | 265,767.81 | 690,165.66 |
| 01-Jul-19 | 506 | 86 | 3.15% | 1,735,375.27 | 1,438,650.08 |
| 01-Aug-19 | 1543 | 109 | 3.99% | 5,104,247.97 | 3,658,208.67 |
| 01-Sep-19 | 1568 | 89 | 3.26% | 5,287,330.59 | 5,960,175.49 |
| 01-Oct-19 | 1639 | 72 | 2.64% | 5,653,893.71 | 8,404,346.49 |
| 01-Nov-19 | 1677 | 100 | 3.66% | 5,358,573.68 | 10,712,822.79 |
| 01-Dec-19 | 2056 | 101 | 3.70% | 6,827,376.62 | 13,653,751.05 |
| | **25164** | **2732** | **100.00%** | **73,816,412.16** | **17,378,757.49** |

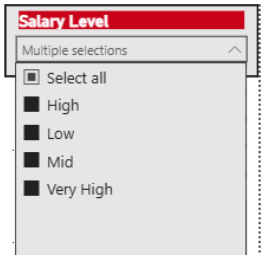
**E.** **Matrix visualization** with the following
   - Rows: start of the month
   - Values: (Total Revenue, Previous Month Revenue, Target Revenue, Revenue Target Achieved)
   - For Revenue Target Achieved column, need to color data with value of 1 to be Green.

| Start of Month | Total revenue | Prev month revenue | Target Revenue | Revenue Target Achieved |
|---|---|---|---|---|
| 01-Nov-18 | 177,124.93 | 200,815.74 | 210,856.52 | 0 |
| 01-Dec-18 | 328,849.50 | 177,124.93 | 185,981.18 | 1 |
| 01-Jan-19 | 218,210.81 | 328,849.50 | 345,291.98 | 0 |
| 01-Feb-19 | 264,432.41 | 218,210.81 | 229,121.35 | 1 |
| 01-Mar-19 | 198,372.75 | 264,432.41 | 277,654.03 | 0 |
| 01-Apr-19 | 295,432.81 | 198,372.75 | 208,291.39 | 1 |
| 01-May-19 | 337,310.18 | 295,432.81 | 310,204.45 | 1 |
| 01-Jun-19 | 265,767.81 | 337,310.18 | 354,175.69 | 0 |
| 01-Jul-19 | 1,735,375.27 | 265,767.81 | 279,056.20 | 1 |
| 01-Aug-19 | 5,104,247.97 | 1,735,375.27 | 1,822,144.04 | 1 |
| | **73,816,412.16** | **66,359,453.90** | **69,677,426.59** | **1** |

F. Make **a Slicer** with the following:

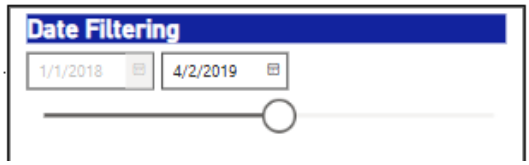
**1- Salary Level slicer**

- Filed: **Salary Level** from Clients Table.
- Enable select all, and enable multi-selection
- Slicer style to be dropdown




**2- Date Slicer**

- Filtering date before choosing date
- Use Header name and format as the picture.

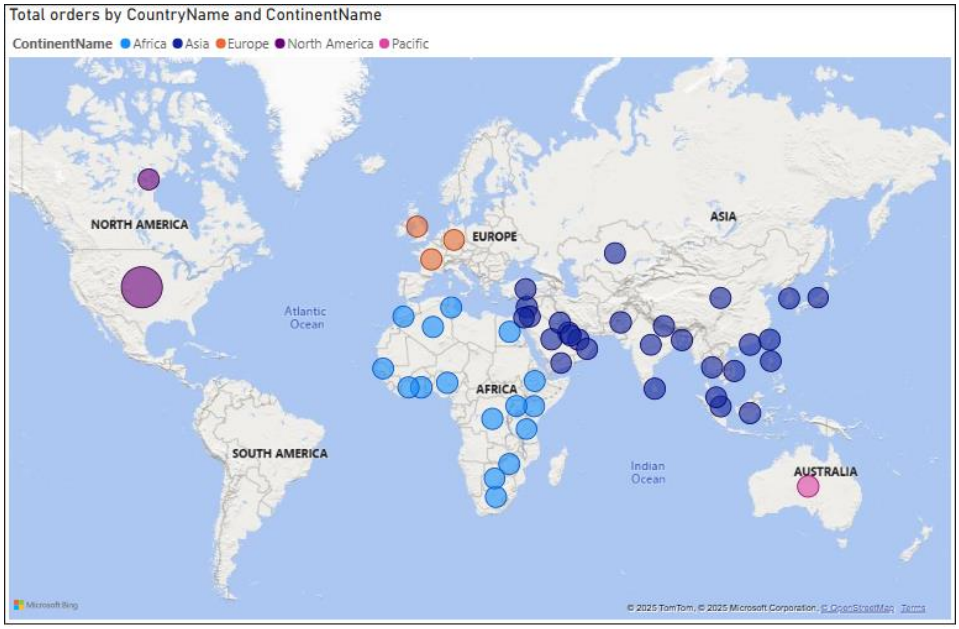



G. **KPI Visuals**

- Create **KPI Visuals comparing current vs. previous month** values for **Start of the week trend**
  1- **Total Orders** (Total Orders vs Previous Months Orders)
  2- **Total Revenue** (Total Revenue vs Previous Months Revenue)
  3- **Total Return** (Total Return vs Previous Months Return)
- Use the same figure titles for each KPI with same color format below for the title.
- **Hint**: for Returns (Goals: Decreasing is positive), (Trend: Low is good)



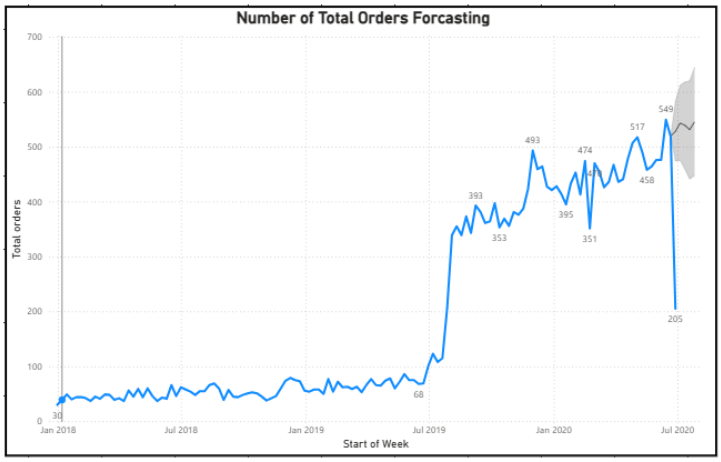
H. Create **a map on the *Map* report page** showing **Total Orders distribution**




## I. Time series Prediction

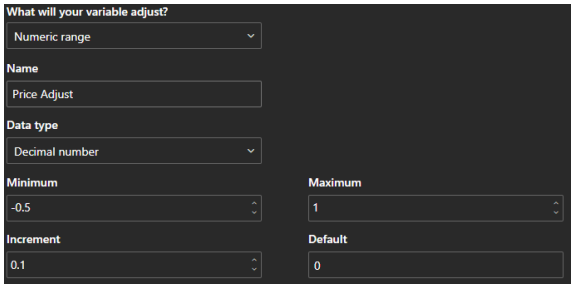
In the Time Series report page, insert Line Chart for "start of the week vs Total Orders" with the below:

- Enable Data Labels.
- Ignore last week
- Forecasting for the next 5 weeks
- Seasonality 14 Weeks
- Figure title: Number of Total Orders Forecasting



## J. What if Scenario

On the Time series page, we need to add a new parameter named Price Adjust with the following details:




- Create a new measure "New Price" = average price * (1+ Price Adjust)
- Create a new measure called Total Adjust Revenue which sum (New Price * OrderQuentity ) from sales table.
- Add the new Adjust Revenue to the Time series plot as below
- Title: The Impact of Price Adjustments on Revenue

